	Ensembl VEP 116.0 `57ea5c5`	fastVEP 0.4.0+6 `5d8004a`	vep-rs 0.4.0 `f642723`	**vepyr** 0.10.0 `3f21520`
Language	Perl	Rust	Rust	Rust, Python
Parallelism	processes	threads	threads	threads
Cache format	Storable, gzip per 1-Mb region	bincode + zstd (transcripts), OSAv2 (annotations)	JSON, gzip optional	Parquet, ZSTD (core and plugins)
Fields filled[a]	80/80	≤37/80	≤71/80 (34/80)[b]	**80/80**
Transcript sets	Ensembl, RefSeq, merged	Ensembl, RefSeq, merged[c]	Ensembl, RefSeq, merged[d]	Ensembl, RefSeq, merged
Declared variant classes	SNV, indel, SV	SNV, indel, SV[e]	SNV, indel, SV[e]	SNV, indel
External annotation sources	yes	yes	yes	yes
Concordance (chr22, merged)	reference	**TODO**	**TODO**	**TODO**
Interfaces	CLI, web, REST	CLI, web, REST	CLI	Python API, SQL, CLI
Evaluation	streaming	streaming	streaming	**lazy, pushdown**
Output	VCF, tab, JSON	VCF, tab, JSON	VCF, tab, JSON, Parquet	Polars DataFrame/LazyFrame, VCF

[a] Of the 80 `CSQ` fields of VEP 116 `--everything` (Ensembl cache), those the tool can fill, counted from its source code and documentation. [b] 34/80 on the cache converted from VEP, used for all concordance results (no known variants, SIFT or PolyPhen). [c] From Ensembl and RefSeq GFF3 files (merged: both, with `SOURCE`); RefSeq-specific fields are not written. [d] Through a cache converted from the VEP RefSeq or merged cache; `SOURCE` and the RefSeq fields are not written. [e] As declared by the authors; SV annotation not fully validated against VEP.